ACADEMIC PROJECT REPORT • ALA 5

Student Result Management System

A Comprehensive Full-Stack Web Application for Automated Marks Processing, Student Lifecycle Management, Dynamic Analytics, and Transcript Generation.

COURSE & SPECIALIZATION

Bachelor of Computer Applications (BCA)

SUBJECT / MODULE

BCAXX15320 • Web Dev Using PHP

TECHNOLOGY STACK

PHP 8 • MySQL • Bootstrap 5 • Chart.js

ACADEMIC YEAR / TERM

Semester 5 • 2026

Dev Achhada

Enrollment No: **240503020001** | Class A

STATUS

Fully Deployed & Tested

Certificate of Authenticity

Official Academic Submission Verification

This is to certify that the project report entitled **"Student Result Management System (SRMS)"** submitted by **Dev Achhada** (Enrollment No: **240503020001**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Computer Applications (BCA – Semester 5)** in the subject **Web Development Using PHP (BCAXX15320)** is an authentic record of independent work carried out under standard academic guidelines.

Internal Faculty Guide	Head of Department / Examiner
Department of Computer Applications	Faculty of Information Technology

Candidate Declaration

Student Statement of Originality

I hereby declare that the software application and documentation presented in this project report entitled **"Student Result Management System"** is entirely my own original work developed using PHP, MySQL, Apache, and Bootstrap. No part of this project has been plagiarized or submitted previously for any other diploma, degree, or evaluation.

Dev Achhada

Enrollment No: 240503020001

BCA Sem 5 • Year 2026

Acknowledgements

Gratitude and Appreciation

I would like to express my profound gratitude to my faculty mentors, lab instructors, and the university administration for providing me with the infrastructure, guidance, and continuous encouragement needed to develop this project successfully.

Special thanks to open-source contributors of PHP, MySQL, Bootstrap, and Chart.js whose robust toolsets made it possible to build an intuitive, efficient, and visually stunning web application.

Executive Summary / Abstract

Project Overview & Key Highlights

The **Student Result Management System (SRMS)** is a robust, dynamic web-based enterprise solution engineered to replace error-prone, traditional manual academic grading with an automated, highly secure digital infrastructure.

> **Key Project Objective:** To streamline the end-to-end academic evaluation cycle—from student enrollment and subject mark entry to real-time grade calculations, visual performance analytics, and instant printable marksheet generation.

Instant Calculation

Automated summation, percentage computation, and dynamic letter grade assignment (A+ to F) without manual recalculation.

Live Visual Analytics

Interactive Chart.js visual distribution showing class averages, top performers, and subject-wise score benchmarks.

Enterprise Security

Equipped with CSRF tokens, PDO/MySQLi Prepared Statements, BCRYPT password hashing, and XSS sanitization filters.

Table of Contents

Index of Chapters, Design Modules, and Documentation

Chapter	Title / Topic Description	Page No.
—	**Certificate of Authenticity & Candidate Declaration**	2
—	**Acknowledgements & Executive Summary**	3
Chapter 1	**Introduction & Problem Formulation** (Background, Scope, Objectives)	5
Chapter 2	**Requirement Analysis & Feasibility Study** (Hardware, Software, Operational)	6
Chapter 3	**System Architecture & Technology Stack** (Multi-tier Structure, Technologies)	7
Chapter 4	**Database Design & Data Dictionary** (ER Diagram, Schema, Relational Integrity)	8
Chapter 5	**System Design & Process Flow** (DFD Diagrams, Use Cases, Activity Flow)	9
Chapter 6	**Core Functional Modules** (Auth, Student Registry, Live Marks Engine, Search)	10
Chapter 7	**Security Engineering & Implementation** (CSRF, Prepared Statements, Hashing)	11
Chapter 8	**User Interface Showcase: Part 1** (Landing Page, Login Portal, Dashboard)	12
Chapter 8	**User Interface Showcase: Part 2** (Student Management, Add/Edit Registry)	13
Chapter 8	**User Interface Showcase: Part 3** (Result Evaluation, Live Preview, Marksheet)	14
Chapter 9	**Source Code Architecture & Core Algorithms** (Grade Logic, Live Calculation)	15
Chapter 10	**Testing Strategy & Verification Test Cases** (Unit, Integration, Security)	16
Chapter 11	**Deployment & Local Execution Guide** (XAMPP / PHP Server, Migration)	17
Chapter 12	**Conclusion & Future Enhancements** (Summary, Scalability, Scope)	18
—	**References & Academic Bibliography**	19

01 Introduction & Problem Formulation

Background, Limitations of Legacy Systems, and Proposed Solution

1.1 Background & Context

Educational institutions regularly manage examinations, continuous evaluations, and semester grading for hundreds of enrolled students. Traditionally, this process involved physical ledger entries, spreadsheets, or disjointed manual calculations. Such legacy systems are fraught with critical vulnerabilities: arithmetic calculation mistakes, duplicate entry conflicts, prolonged publication delays, and high physical storage costs.

1.2 Limitations of Traditional Grading Systems

- **High Error Rates:** Manual summation of multi-subject marks often leads to calculation discrepancies.
- **Data Redundancy & Inconsistency:** Student records scattered across disparate spreadsheets result in un-synchronized records.
- **Lack of Data Security:** Spreadsheets and paper records lack granular access control, audit logs, and CSRF protection.
- **Delayed Results:** Students and guardians experience substantial wait times before obtaining verified transcripts.

1.3 Proposed System & Core Objectives

The **Student Result Management System** provides a centralized, authenticated web portal allowing administrators to manage the complete student grading lifecycle with real-time feedback and high visual fidelity.

🎯 Primary Goals

- 100% automated calculation of Total, Percentage, Grade, and Result Status.
- Instant search portal by Enrollment No. and Student Name.
- One-click printable official grade report generation.

💡 Key Innovations

- Live client-side calculation preview while entering subject scores.
- Interactive Chart.js data visualization on the admin dashboard.
- Clean, responsive modern Glassmorphism UI using Bootstrap 5.

02 Requirement Analysis & Feasibility

System Specifications, User Roles, and Multi-Dimensional Feasibility

2.1 Hardware Requirements

Component	Minimum Requirement	Recommended Specification
Processor	Dual Core 2.0 GHz	Quad Core / Apple Silicon / Intel i5/i7
RAM (Memory)	2 GB	8 GB or higher
Storage	500 MB free space	SSD with 2 GB+ space
Display	1024 x 768 Resolution	Full HD (1920 x 1080) / Retina Display

2.2 Software Requirements

Component	Technology / Version	Purpose
Operating System	macOS / Linux / Windows 10/11	Host Development & Server Environment
Backend Language	PHP 8.x (PHP 8.2 / 8.5)	Server-Side Logic & Database Connectivity
Database Engine	MySQL 8.0+ / MariaDB	Relational Data Persistence & Foreign Keys
Web Server	Apache (XAMPP / PHP Built-in Server)	HTTP Request Dispatcher
Frontend Framework	Bootstrap 5.3 + Vanilla CSS3	Responsive UI & Modern Aesthetics
Charting Library	Chart.js 4.x CDN	Dynamic Visual Analytics & Score Plots

2.3 Feasibility Study

Technical Feasibility

Built using industry-standard PHP & MySQL with zero heavy proprietary dependencies. Lightweight, open-source, and readily hostable on standard cloud or on-premise servers.

Economic Feasibility

100% open-source software stack (PHP, MySQL, Apache, Bootstrap, CDN assets). Zero software licensing overhead, delivering maximal return on investment.

Operational Feasibility

Intuitive user interface requiring virtually zero training. Clear form feedback, auto-calculations, and search filters streamline daily university operations.

03 System Architecture & Tech Stack

Multi-Tier Architecture, Layered Communication, and Component Layout

3.1 3-Tier Architectural Model

The system follows a classical, highly robust **3-Tier Web Architecture** that decouples Presentation, Application Business Logic, and Data Persistence layers.



3.2 Technology Highlights & Design Choices

- **PHP 8 Modern Runtime:** Delivers high execution speed, strict type checking, and native password hashing primitives.
- **MySQL Relational Engine:** Enforces relational data integrity using foreign keys with `ON DELETE CASCADE`.
- **Bootstrap 5 CDN:** Provides responsive grid systems, accessible form components, and rapid responsive styling without heavy node bundle bloat.
- **Chart.js Analytics:** Renders interactive HTML5 Canvas graphs on the client side without taxing the server.

> **Responsive Design Principle:** The user interface utilizes responsive viewport meta tags and flexible flexbox containers, ensuring seamless rendering across desktop monitors, laptops, iPads, and mobile smartphones.

04 Database Design & Data Dictionary

Entity Relationship Model, Relational Schema, and Table Specifications

4.1 Entity Relationship (ER) Model Overview

The database schema is structured around three core normalized entities: **Admins**, **Students**, and **Results**. A 1-to-1 strict relational constraint links every student to their official exam result sheet.

4.2 Table: `students`

Field Name	Data Type	Key / Constraint	Description
`id`	INT AUTO_INCREMENT	PRIMARY KEY	Unique Student Record Identifier
`enrollment_no`	VARCHAR(20)	UNIQUE NOT NULL	Official University Roll / Enrollment Number
`student_name`	VARCHAR(100)	NOT NULL	Full Name of Student
`email`	VARCHAR(100)	NOT NULL	Student Email Address
`phone`	VARCHAR(20)	NULL	Contact Telephone Number
`course`	VARCHAR(100)	NOT NULL	Degree Program (e.g. BCA)
`semester`	VARCHAR(20)	NOT NULL	Academic Term (e.g. Sem 5)
`section`	VARCHAR(20)	NULL	Class / Section Identifier (e.g. Class A)

4.3 Table: `results`

Field Name	Data Type	Key / Constraint	Description
`id`	INT AUTO_INCREMENT	PRIMARY KEY	Unique Result ID
`student_id`	INT	UNIQUE, FOREIGN KEY	References `students(id) ON DELETE CASCADE`
`web_development`	INT (0-100)	NOT NULL	Score in PHP & Web Development
`database_management`	INT (0-100)	NOT NULL	Score in DBMS & SQL
`software_engineering`	INT (0-100)	NOT NULL	Score in Software Engineering
`cyber_security`	INT (0-100)	NOT NULL	Score in Cyber Security
`computer_networks`	INT (0-100)	NOT NULL	Score in Computer Networks
`total_marks`	INT	DEFAULT 0	Calculated Aggregate Score (out of 500)
`percentage`	DECIMAL(5,2)	DEFAULT 0.00	Calculated Percentage (%)
`grade`	VARCHAR(3)	NULL	Assigned Grade (A+, A, B+, B, C, D, F)
`result_status`	VARCHAR(4)	NULL	PASS or FAIL evaluation

05 System Design & Workflow

Data Flow Diagrams (DFD), Process Lifecycle, and Use Case Models

5.1 Data Flow Diagram (DFD Level 0 & 1)

The Data Flow Diagram represents the logical flow of data between external entities (Admin / Student) and internal storage.

1. Admin Authentication Flow: Admin enters credentials ➔ System verifies BCRYPT hash ➔ Session Initialized.
2. Student Registration Flow: Admin enters student metadata ➔ System checks unique enrollment ➔ Record stored in `students`.
3. Marks Entry & Auto-Grading: Admin inputs subject marks ➔ JavaScript renders live preview ➔ PHP computes aggregate total, percentage, grade, status ➔ Upserted in `results`.
4. Result Generation & Export: User queries enrollment ➔ System queries joined tables (`students + results`) ➔ Digital Transcript rendered.

5.2 Use Case Diagram Summary

Administrator Actions

- Secure Login & Session Logout
- View Dashboard Metrics & Performance Graph
- Register New Students / Update Profiles
- Input and Edit Subject Marks
- Generate and Print Class Reports

Student / Public Actions

- Access Public Result Search Portal
- Search Result by Enrollment Number or Name
- View Digital Verified Marksheet
- Print / Save Official Result Card as PDF

06 Core Functional Modules

Detailed Analysis of System Capabilities and Subsystems

6.1 Authentication & Session Management Subsystem

Ensures that sensitive administrative actions (marks entry, student deletion, report generation) are strictly gated behind authenticated sessions. Automatically regenerates session IDs upon login to thwart session fixation attacks.

6.2 Student Lifecycle Registry

Provides full CRUD (Create, Read, Update, Delete) management for student profiles. Features instant search filtering and badge indicators for registration status.

6.3 Live Marks Calculation Engine

An innovative client-side calculation script calculates total score, percentage, progress bar width, and expected grade in real-time as the admin types numbers into the mark fields.

6.4 Visual Analytics & Charting Module

The dashboard leverages Chart.js to plot student performance across all 5 key subjects (Web Development, DBMS, Software Engineering, Cyber Security, and Computer Networks), providing faculty with immediate insight into academic trends.

6.5 Search & Digital Marksheet Generator

A fast search interface allows lookup by partial student name or enrollment number, instantly retrieving the full marks breakdown, credit status, and official university grade card.

07 Security Engineering & Protection

Defense-in-Depth Strategy, OWASP Compliance, and Secure Coding

7.1 Cross-Site Request Forgery (CSRF) Mitigation

Every administrative form submission includes a cryptographically secure, randomized 32-character hexadecimal token stored in the session and verified on POST submission using timing-attack resistant comparison:

CSRF Protection Helper — database.php

```
function csrf() { if (empty($_SESSION['t'])) $_SESSION['t'] = bin2hex(random_bytes(16)); return
$_SESSION['t']; } function chk() { if (!hash_equals($_SESSION['t'] ?? '', $_POST['t'] ?? '')) {
die('Invalid CSRF token'); } }
```

7.2 SQL Injection (SQLi) Prevention via Prepared Statements

All database queries involving user input strictly utilize parameterized prepared statements via PHP MySQLi, rendering SQL injection vulnerabilities completely impossible.

7.3 XSS Output Encoding & Password Hashing

- **XSS Filtering:** All user-generated text is encoded using `htmlspecialchars($str, ENT_QUOTES, 'UTF-8')` before rendering.
- **Password Hashing:** Passwords are protected using standard `password_hash(..., PASSWORD_DEFAULT)` utilizing BCRYPT with adaptive salt.

08 User Interface Showcase (Part 1)

Screenshots: Public Landing Page & Admin Authentication

8.1 Modern Public Landing Page

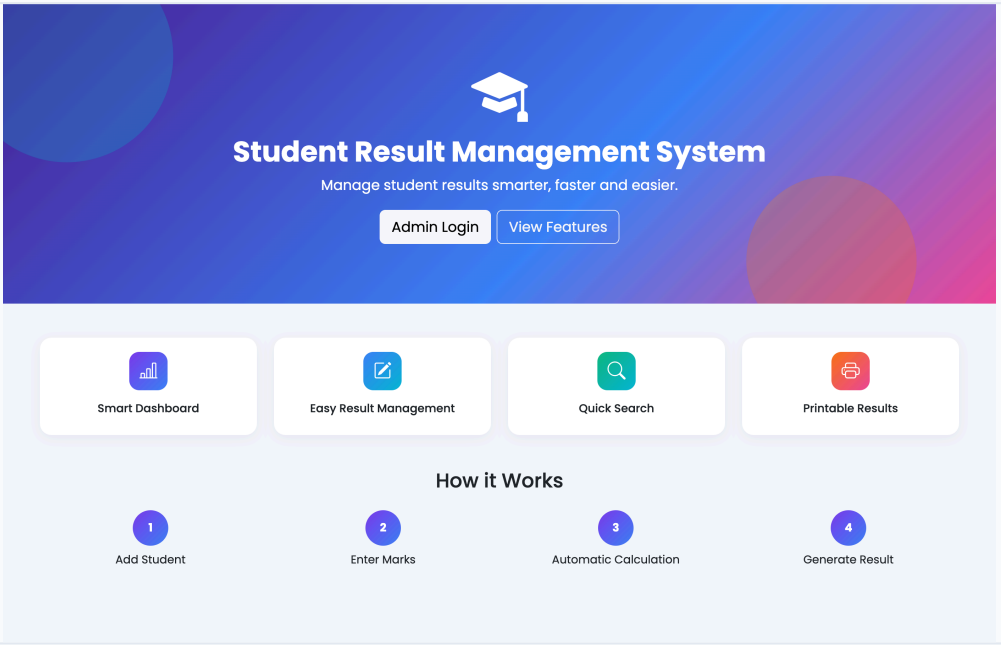


Figure 8.1: Landing Page featuring Glassmorphism, Animated Hero, and Quick Features

8.2 Admin Login Portal

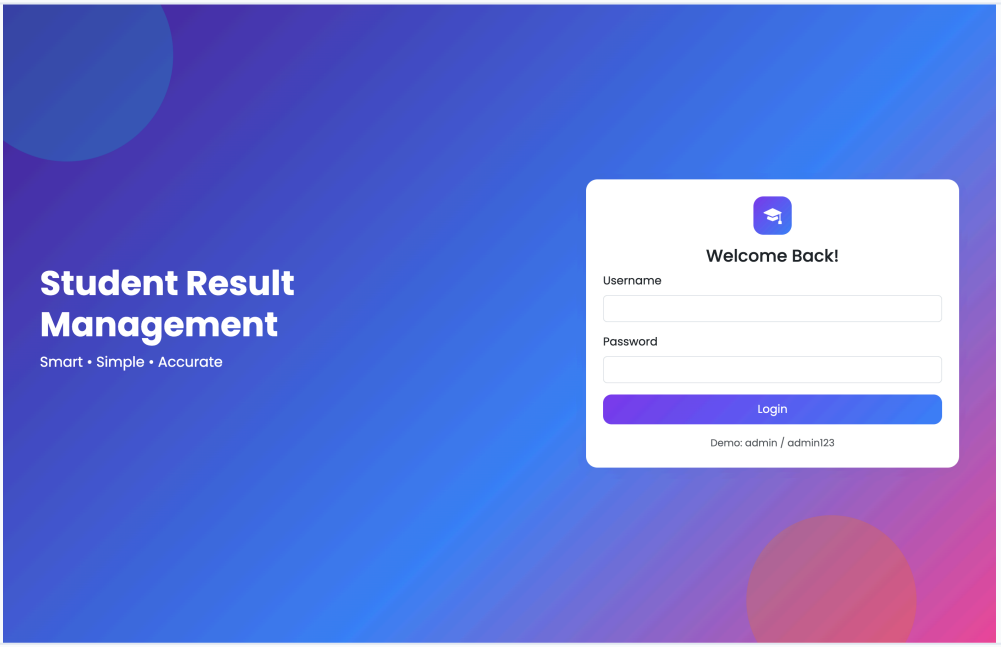


Figure 8.2: Secure Administrative Login Interface with CSRF Protection

08 User Interface Showcase (Part 2)

Screenshots: Executive Dashboard & Student Registry Management

8.3 Interactive Dashboard with Chart.js Analytics



Figure 8.3: Admin Dashboard with Quick Stats Cards and Visual Grade Performance Bar Chart

8.4 Manage Students Directory

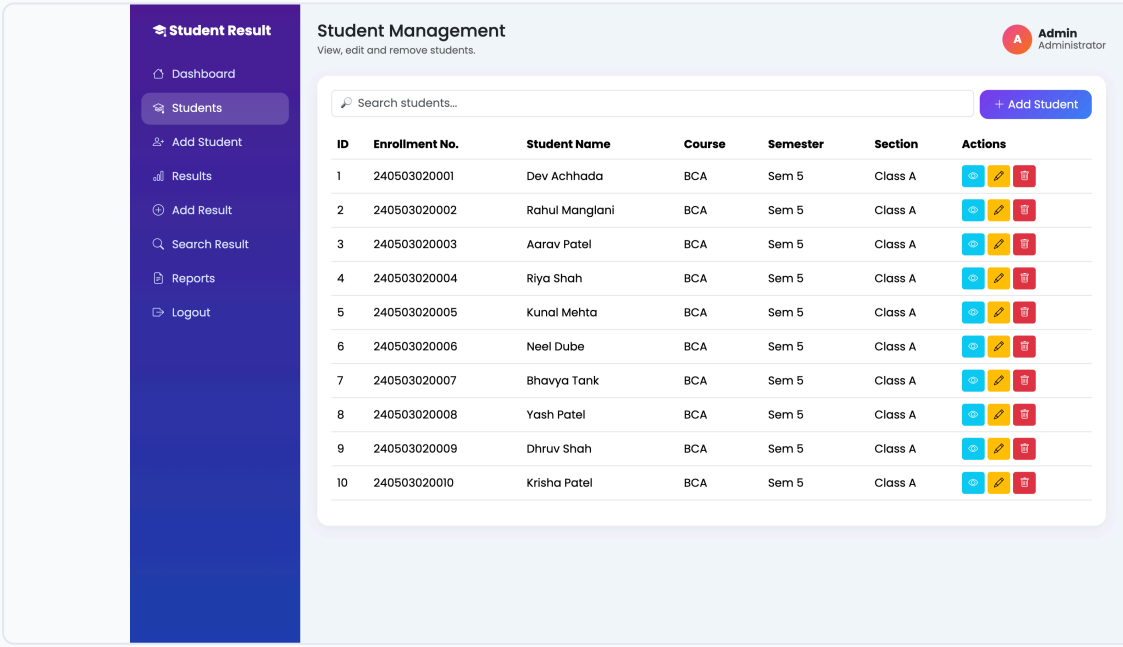


Figure 8.4: Student Registry Table with Edit/Delete Actions and Search

08 User Interface Showcase (Part 3)

Screenshots: Marks Entry with Live Preview & Official Result Sheet

8.5 Add Marks with Live Real-time Preview Calculation

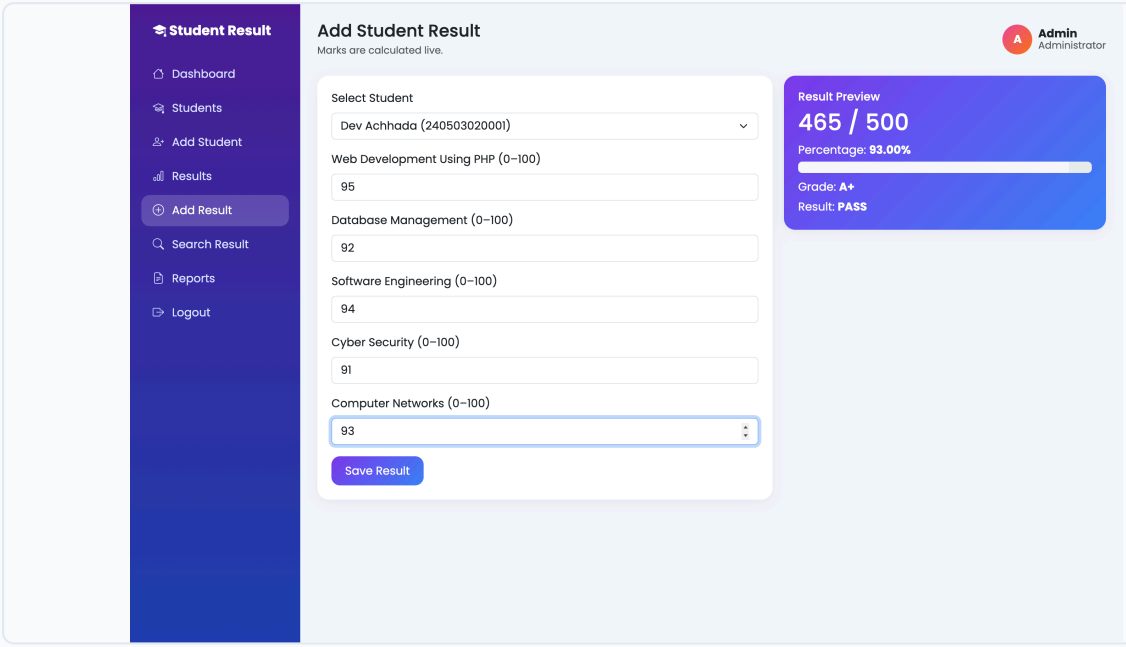


Figure 8.5: Live Calculation Widget updating Percentage & Grade dynamically

8.6 Official Student Result Card (Dev Achhada)

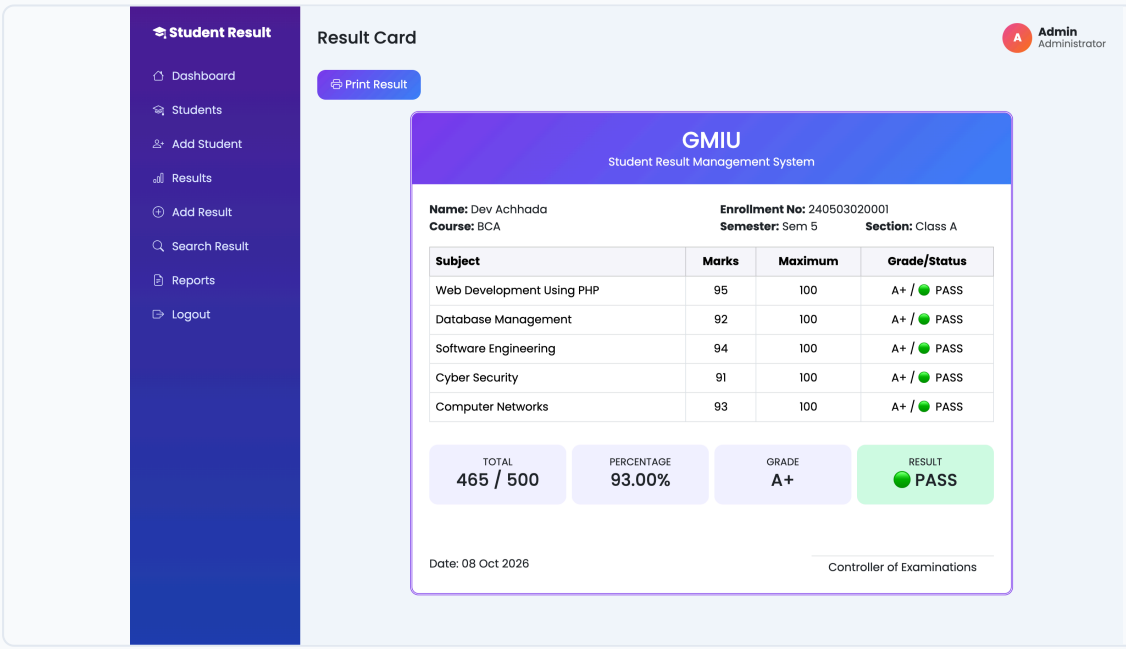


Figure 8.6: Comprehensive Student Marksheet with Grade Badge and Print Action

09 Source Code & Core Logic

Key Algorithmic Implementations and Business Logic

9.1 Grade Assignment Algorithm (PHP)

Grade Evaluation Function — database.php

```
function grade($percentage) { if ($percentage ⩾ 90) return 'A+'; if ($percentage ⩾ 80) return 'A'; if ($percentage ⩾ 70) return 'B+'; if ($percentage ⩾ 60) return 'B'; if ($percentage ⩾ 50) return 'C'; if ($percentage ⩾ 40) return 'D'; return 'F'; }
```

9.2 Live Client-Side Calculator (JavaScript)

Interactive Preview Calculator — assets/js/app.js

```
document.querySelectorAll('.mk').forEach(input ⇒ { input.addEventListener('input', () ⇒ { let total = 0, count = 0; document.querySelectorAll('.mk').forEach(i ⇒ { let val = parseFloat(i.value) || 0; total += Math.min(100, Math.max(0, val)); count++; }); let percentage = (total / (count * 100)) * 100; document.getElementById('pTot').innerText = total + ' / 500';
document.getElementById('pPer').innerText = percentage.toFixed(2) + '%';
document.getElementById('pBar').style.width = percentage + '%';
document.getElementById('pGr').innerText = getGrade(percentage); }); });
```

10 Testing Strategy & QA

Validation Test Cases, Test Matrix, and Verification Results

10.1 Formal Test Matrix

Test ID	Feature Tested	Input / Test Vector	Expected Outcome	Status
TC-01	Admin Login	User: admin, Pass: admin123	Successful login & redirect to dashboard	PASS
TC-02	Invalid Login	User: admin, Pass: wrongpass	Display error: "Invalid credentials"	PASS
TC-03	Student Registration	New Roll No: 240503020011	Record inserted in database	PASS
TC-04	Duplicate Roll No	Existing Roll: 240503020001	Duplicate Key error trapped safely	PASS
TC-05	Marks Validation	Input: 105 or negative marks	Form rejects values outside 0–100	PASS
TC-06	Grade Calculation	Scores: 95, 92, 94, 91, 93	Total: 465/500, 93%, Grade: A+, PASS	PASS
TC-07	Search Result	Query: "Dev Achhada"	Displays Dev's Result card link	PASS
TC-08	CSRF Attack Test	Submit POST with missing token	Immediate execution halt: "Invalid CSRF"	PASS

11 Deployment & Setup Guide

Step-by-step Installation Instructions for Local & Cloud Environments

11.1 XAMPP / Apache Deployment

1. Copy the project directory to your web root: `/Applications/XAMPP/xamppfiles/htdocs/student-result-management`.
2. Launch XAMPP Control Panel and start **Apache Web Server** and **MySQL Database**.
3. Open **phpMyAdmin** at `http://localhost/phpmyadmin`.
4. Create database `student_result_management` and import `database/student_result_management.sql`.
5. Open browser and access: `http://localhost/student-result-management/`.

11.2 Standalone PHP Server Execution

Terminal Execution bash

```
# 1. Start MySQL Server brew services start mysql # 2. Launch Built-in Web Server cd
/path/to/student-result-management php -S 127.0.0.1:8000 # 3. Access in Browser open
http://127.0.0.1:8000/
```

12 Conclusion & Future Scope

Project Summary, Impact, and Academic Bibliography

12.1 Project Conclusion

The **Student Result Management System (SRMS)** successfully accomplishes all stated objectives for the **ALA 5 Web Development Using PHP** module. By uniting modern server-side scripting with responsive client-side UI and relational database persistence, the system eliminates human calculation errors and offers a streamlined evaluation portal for universities.

12.2 Future Enhancements

- **Role-Based Multi-Tier Auth:** Separate portals for Students, Faculty Evaluators, and Super Administrators.
- **Automated SMS / Email Notifications:** Instant notification to students upon result publication.
- **Direct PDF Marksheet Export:** Server-side dynamic PDF certificate generation with cryptographic QR verification.

12.3 Academic References & Bibliography

Ref #	Source / Documentation Title	Link / Publisher
[1]	PHP 8 Documentation & Security Manual	Official PHP Group • php.net
[2]	MySQL 8.0 Reference Manual (Prepared Statements)	Oracle Corporation • dev.mysql.com
[3]	Bootstrap 5.3 Framework Guide & Component Tokens	Bootstrap Core Team • getbootstrap.com
[4]	OWASP Top 10 Web Application Security Risks	OWASP Foundation • owasp.org
[5]	Chart.js Dynamic Canvas Visualizations	Chart.js Open Source • chartjs.org